For flexible encoders the contrastive loss stops mattering; the score matters only through its tails

Thesis

For flexible encoders and infinite data, the choice of contrastive loss (one-way, two-way or joint) stops mattering: all three aim at the same target and can all reach it. What still matters is the choice of *score*, and it matters only through the score's *tail behaviour*:

- a score that is **bounded above** (L^2, normalised cosine) defines a valid model for every data distribution, and the forecast built from a bank of real futures converges to that model as the bank grows;
- a score that **grows without bound** in the future (the unnormalised dot product) has no valid model on heavy-tailed data, and its bank forecast degenerates as the bank grows, putting all its weight on a single future.

Experiments confirm both halves, with one refinement: a ceiling is necessary but not sufficient in practice. Training never sees the far tails, so the score must also not *reward* extreme futures, which L^2 and normalised cosine do not.

"Valid model" has a precise meaning here: the normalising constant of the model's forecast, $Z(u)$ below, is finite, so the forecast is a probability distribution and the loss has a finite value. When it is infinite, the object the loss is trying to learn does not exist.

The note builds the thesis in four steps. Section 1 sets out the framework of Baptista, Stuart & Tran [1] that everything rests on. Section 2 collects what three rounds of experiments showed when we took that framework from its Gaussian, linear setting to time series. Section 3 proves the first half of the thesis (the loss stops mattering) and Section 4 the second (the tails decide). Section 5 reports two new experiments that test the second half directly. Section 6 explains what changes with finite data, and Section 7 places the results in the literature.

1 The framework: a contrastive model is a forecast

Data and score. The past is u, the future is v, the data distribution is $\mu(du, dv)$, and its marginals are μ_u, μ_v. We assume μ and $\mu_u \otimes \mu_v$ are mutually absolutely continuous and write

$$r(u, v) = \frac{d\mu}{d(\mu_u \otimes \mu_v)}(u, v) = \frac{p(v \mid u)}{p(v)}.$$

So $\log r$ is the pointwise mutual information: how much more likely this future is given this past than it is in general. Its average is the mutual information $I(u; v) = \mathbb{E}_\mu \log r$.

A contrastive model is a *score* $s(u, v)$, built from a past encoder f, a future encoder g and a scale $\kappa > 0$ (the learned logit scale). We use three scores:

$$s_{\mathrm{dot}} = \kappa \, \langle f(u), g(v) \rangle, \qquad s_{\mathrm{ncos}} = \kappa \left\langle \tfrac{f(u)}{|f(u)|}, \tfrac{g(v)}{|g(v)|} \right\rangle, \qquad s_{L^2} = -\tfrac{\kappa}{2} \, |f(u) - g(v)|^2.$$

s_{dot} is the paper's unnormalised "cosine" score, s_{ncos} is real CLIP's normalised cosine, and s_{L^2} is the paper's L^2 score. Note that $s_{L^2} \le 0$ and $|s_{\mathrm{ncos}}| \le \kappa$, while s_{dot} is unbounded whenever g is.

The model. The paper's central observation (its eqs. (13)–(16)) is that a score defines a probability model by tilting the product of the marginals:

$$\nu(du, dv) = \frac{e^{s(u,v)}}{Z} \, \mu_u(du) \, \mu_v(dv), \qquad \nu_{v|u}(dv \mid u) = \frac{e^{s(u,v)}}{Z(u)} \, \mu_v(dv),$$

$$Z = \iint e^s \, d\mu_u \, d\mu_v, \qquad Z(u) = \int e^{s(u,v)} \, \mu_v(dv).$$

In words: the forecast for a past u takes the distribution of all futures and reweights each one by e^{score}.

The losses. The CLIP-style losses are, up to constants, divergences between the data and this model (the paper's Thm 3.2, Thm 3.8 and Remark 3.12):

$$\begin{aligned} J_{v|u}(s) &= \mathbb{E}_u \, D_{\mathrm{KL}}\big(\mu_{v|u}(\cdot \mid u) \,\|\, \nu_{v|u}(\cdot \mid u)\big) && \text{one-way (the forecasting direction),} \\ J_2(s) &= \lambda_u J_{v|u}(s) + \lambda_v J_{u|v}(s), \quad \lambda_u, \lambda_v > 0 && \text{two-way (standard CLIP),} \\ J_{\mathrm{joint}}(s) &= D_{\mathrm{KL}}(\mu \,\|\, \nu) && \text{joint (one softmax over all pairs).} \end{aligned}$$

The forecast we actually compute. The paper's eq. (31) turns the model into a forecast using a finite *bank* $v_1, \ldots, v_N$ of real futures:

$$\nu^N_{v|u}(\cdot \mid u) = \sum_{j=1}^{N} w_j \, \delta_{v_j}, \qquad w_j = \frac{e^{s(u,v_j)}}{\sum_k e^{s(u,v_k)}}, \qquad \mathrm{ESS} = \Big(\sum_j w_j^2 \Big)^{-1}.$$

This is self-normalised importance sampling with proposal μ_v. ESS, the effective sample size, counts how many bank futures the forecast really uses. ESS ≈ 1 means the forecast has collapsed onto a single future.

The paper's predictions. With Gaussian data and linear encoders, the paper derives every minimiser in closed form. Four predictions matter here:

P1	The dot product gets the forecast's average right but its spread wrong: the spread is stuck at "knowing nothing" (Cor. 5.2).
P2	L^2 trained one-way gets both average and spread right (Thm 5.3, Cor. 5.4).
P3	For the dot product, one-way and two-way give the same answer (Thm 5.1).
P4	The joint loss shrinks the forecast towards the average, so it is a worse forecaster than the one-way loss (Thm 5.6, Cor. 5.7).

2 What the experiments showed

We tested these predictions in three rounds, relaxing one of the paper's assumptions at a time.

Protocol. Every contrastive model is scored only through its own score, by building $\nu^N_{v|u}$ from a bank of real futures that were not used in training. Forecasts are scored with the *energy score* [16], a proper scoring rule for multivariate probabilistic forecasts that rewards both accuracy and honest uncertainty. On synthetic data each domain has an exact "perfect forecaster" (oracle), and we report the energy score as a percentage above the oracle's (0% = perfect). On real data we report *skill*, the fractional reduction in energy score against a reference forecast (higher is better). Each synthetic setting uses 200k training pairs, a bank of 50k held-out futures, 500 test cases and 2 seeds, so differences under about 1% are noise. Past and future are windows of 50 points of a smooth Gaussian process observed with noise.

Domains. Each synthetic domain has a dial whose zero point is the Gaussian process, which is the paper's setting:

- **heavy tails**: an elliptical Student-t process with ν degrees of freedom ($\nu = \infty$ is the Gaussian process, $\nu = 3$ the heaviest we used);
- **bent GP**: a monotone warp of the Gaussian process (skewed futures, nonlinear relation between past and future);
- **double-well**: a stochastic process that jumps between two levels, so the future can have two separate likely outcomes.

The real data are one year (2020) of 10-minute readings from the Jena weather station. We forecast the next 6 h of temperature from the last day of six variable groups, and the next 6 h of rain from past pressure, wind and moisture.

Finding 1: with linear encoders, both the loss and the score matter. On the Gaussian process the dot product is +5.6% above perfect whichever direction it is trained in (P3), and +9.8% when trained jointly (P4). Its 90% bands catch the truth 98% of the time on the hardest third of cases (P1: too wide). L^2 is at +0.0% in all three variants. Linear normalised cosine is poor everywhere (+12–23%), because it sees the past only through its direction and loses its level.

Finding 2: with MLP encoders, the loss variant mostly stops mattering. With two-layer MLP encoders (width 256, plus a linear shortcut), the spread between the variants collapses:

domain	encoder	dot 1-way	dot 2-way	dot joint	spread
Gaussian process	linear	+5.6	+5.6	+9.8	4.2
	MLP	+2.7	+1.4	+0.3	2.4
bent GP (full warp)	linear	+37.4	+40.0	+25.2	14.8
	MLP	+5.9	+3.9	+0.6	5.3
double-well ($\lambda = 1.5$)	linear	+1.3	+2.6	+2.7	1.4
	MLP	+0.6	+0.3	+0.5	0.3

P1 also fades: the dot product's coverage on the hardest cases drops from 98% to 89%. The best models (L^2 joint, normalised cosine) are within about 1% of perfect on every domain. The differences that remain are small and consistently favour the joint loss; we return to them in Section 6.

Finding 3: every catastrophic failure is an unbounded score on heavy tails.

heavy tails, $\nu = 3$	dot 2-way	normalised cosine	L^2 joint
linear	$\sim 10\times$ worse than perfect	+23.0	+4.5
MLP	+23.3	+0.9	+0.7

The linear dot product's forecasts put essentially all their weight on single bank futures (minimum ESS 1.0). The MLP does not rescue it: MLP dot products are still +7 to +23% above perfect and still collapse onto single futures for some inputs. On rain, which is mostly zero with occasional bursts, the unnormalised dot product scored 1.4–2.4 times worse than climatology, while L^2 and normalised cosine kept the shape "mostly dry, sometimes a burst". Normalisation helps exactly where tails are heavy, and only there.

Finding 4: with real, limited data, the loss matters again. On weather, MLPs over-fit within a few hundred to a few thousand steps. There, joint L^2 clearly beat one-way L^2 (temperature skill +0.28 vs +0.15 against persistence). Section 6 explains why this does not contradict the thesis, which is a statement about infinite data.

3 Why the loss stops mattering

All three losses aim at the same object, the log density ratio. They differ only in which part of it they leave free.

Proposition 1 (Common target)**.** *For any score s:*

(i) $J_{v|u}(s) = 0 \iff s(u,v) = \log r(u,v) + c(u)$ *for some function c;*

(ii) $J_{u|v}(s) = 0 \iff s(u,v) = \log r(u,v) + c'(v)$ *for some function c';*

(iii) $J_2(s) = 0 \iff J_{\text{joint}}(s) = 0 \iff s = \log r + \text{const.}$

Definition 2. A score family $\mathcal{S}$ is *well-specified* for a loss if that loss can be driven to 0 on $\mathcal{S}$. By Proposition 1, that means $\mathcal{S}$ contains some $\log r + c(u)$ for the one-way loss, and $\log r + \text{const}$ for the two-way and joint losses.

Corollary 3 (Agreement)**.** *If $\mathcal{S}$ contains $\log r + \text{const}$, then the one-way, two-way (any $\lambda_u, \lambda_v > 0$) and joint losses all have minimisers that give exactly the true forecast, $\nu_{v|u} = \mu_{v|u}$. The loss variants cannot be told apart at the population level.*

So a difference between loss variants is a symptom of a score family that cannot represent $\log r$. The converse fails: P3 is a case where the family cannot represent $\log r$, yet two variants still agree.

Flexible encoders make the family well-specified. With biases, $\langle f(u), g(v)\rangle$ contains terms $c(u)+d(v)$. Sums $\sum_k f_k(u)g_k(v)$ of products of network features are dense in the continuous functions on compact sets, by Stone–Weierstrass. Stuart & Wolf [2] have made this precise for two-tower CLIP: with flexible enough encoders and a learnable temperature, the model can approximate any joint distribution on a compact domain to arbitrary accuracy in KL. Combined with Corollary 3, this is the first half of the thesis: once the encoders are flexible, the loss variants converge to the same forecast. Finding 2 is that convergence, seen at finite data.

3.1 What happens when the family is too rigid: the Gaussian case

The Gaussian case shows how the loss variants come apart when the family is not flexible, and recovers P1, P3 and P4. Let μ be a centred Gaussian with covariance blocks C_{uu}, C_{uv}, C_{vv}, joint precision P and $D = \mathrm{blockdiag}(C_{uu}, C_{vv})$. Then

$$\log r(u,v) = -\tfrac{1}{2}u^\top(P_{uu} - C_{uu}^{-1})u \;-\; u^\top P_{uv} v \;-\; \tfrac{1}{2}v^\top(P_{vv} - C_{vv}^{-1})v + \text{const.}$$

There are three ingredients: a past–future interaction, which sets the forecast's average; a future-only term, which sets its spread; and a past-only term. With linear encoders every score is $s = -\frac{1}{2}u^\top Ru + u^\top Av - \frac{1}{2}v^\top Sv$, and the families differ in which ingredients they can set:

family	allowed (R, A, S)	well-specified for
dot product	$R = S = 0$	nothing (if $C_{uv} \neq 0$)
L^2	$\left(\begin{smallmatrix} R & -A \\ -A^\top & S \end{smallmatrix}\right) = [G, -H]^\top[G, -H] \succeq 0$	one-way only
dot + free quadratics	R, S symmetric, A free	every loss

- **P1.** The dot product has no future-only term, so its forecast's covariance is C_{vv} whatever A is: the spread is stuck at climatology.
- **P4.** The joint loss also needs the past-only term, which the dot product lacks. On top of that, the joint normaliser is finite only if $\|C_{uu}^{1/2} A C_{vv}^{1/2}\| < 1$, which is where the paper's shrinkage function h comes from. Our numerical minimiser reproduces h exactly.
- **P2.** One-way L^2 is well-specified, because $S \succeq 0$ can match the future-only term.

Proposition 4 (Linear L^2 is too rigid for the two-way and joint losses)**.** *Under linear L^2, every model joint covariance satisfies $(D^{-1} + Q)^{-1} \preceq D$, with $Q = [G, -H]^\top[G, -H] \succeq 0$. The data satisfy $C \not\preceq D$ whenever $C_{uv} \neq 0$. So the two-way and joint losses cannot reach zero. In words: an L^2 model can couple past and future only by shrinking both marginals, and real data couple them without shrinking.*

Proposition 5 (Scalar closed form)**.** *For unit-variance scalars with correlation $\rho \in (0, 1)$, the joint-L^2 optimum forecasts $\mathcal{N}(\frac{\rho}{2-\rho}u,\ \frac{2(1-\rho)}{2-\rho})$ against the truth $\mathcal{N}(\rho u, 1-\rho^2)$. The average is shrunk and the forecast is over-confident. The fraction of the information $I(u;v)$ it loses tends to $3/8$ as $\rho \to 0$ and to 0 as $\rho \to 1$.*

		dot product			L^2			dot + quad.
setting	$I(u;v)$	1-way	2-way	joint	1-way	2-way	joint	all three
$\rho = 0.1$	0.005	0.005	0.005	0.005	0	0.294	0.333	0
$\rho = 0.3$	0.047	0.046	0.046	0.051	0	0.219	0.251	0
$\rho = 0.5$	0.144	0.131	0.131	0.157	0	0.148	0.170	0
$\rho = 0.7$	0.337	0.272	0.272	0.323	0	0.081	0.092	0
$\rho = 0.9$	0.830	0.512	0.512	0.571	0	0.021	0.023	0
our Gaussian process	1.627	0.704	0.704	0.746	0	0.003	0.003	0

Table 1: Fraction of the information $I(u;v)$ (in nats) lost by the forecast at each loss's population optimum: 0 is perfect, 1 is climatology. Exact, from Gaussian KL divergences, with no training or sampling. The last row uses the exact covariances of our synthetic Gaussian process.

Table 1 makes the picture quantitative. Two points stand out.

- **Each rigid family fails in its own regime.** The dot product's error is in the spread, which matters most when dependence is strong. L^2's two-way/joint error is the coupling-by-shrinking constraint, which matters most when dependence is weak. Our Gaussian process is strongly dependent ($I = 1.63$ nats), so two-way and joint L^2 lose only 0.3% of the information there. That is why Finding 1 showed all three L^2 variants at +0.0%.
- **Adding the missing ingredients removes every difference.** Giving the dot product free past-only and future-only quadratic terms makes all three losses exact (last column). This is the flexibility argument in miniature: complete the family and the loss stops mattering.

4 Why the score still matters: its tails

Flexibility is about representing $\log r$ well where the data are. It says nothing about how the score behaves far out, and that is where the second half of the thesis lives. Stuart & Wolf's extension of their approximation result to unbounded domains already needs the score to be *bounded* outside a compact set, with an error term that grows like e^B in that bound [2]. We show what goes wrong without it.

Definition 6. A score s is *admissible* for μ if $Z(u) < \infty$ for μ_u-almost every u. Then the forecast $\nu_{v|u}$ exists and $J_{v|u}(s)$ is finite. Otherwise $J_{v|u}(s) = +\infty$ and there is no model to learn.

Proposition 7 (Bounded above ⇒ admissible for every distribution)**.** *If $s \le c$, then $Z(u) \le e^c$ and $Z \le e^c$ for every μ. So L^2 ($c = 0$) and normalised cosine ($c = \kappa$) are admissible for every data distribution, with any encoders.*

Proposition 8 (A score linear in the future needs light tails)**.** *If $s(u, v) = \langle a(u), Hv\rangle + b(u)$, then $Z(u) = e^{b(u)} M_{\mu_v}(H^\top a(u))$, where M_{μ_v} is the moment generating function of the futures. For a multivariate Student-t (any degrees of freedom), $M_{\mu_v}(w) = \infty$ for every $w \neq 0$. So the linear dot product is admissible only if it ignores the past: every informative parameter has infinite loss.*

Proposition 9 (ReLU networks do not rescue the dot product)**.** *Let g be a ReLU network and g_∞ the same network with its biases removed. If $\langle f(u), g_\infty(v_0)\rangle > 0$ for some direction v_0 and the futures are elliptical Student-t, then $Z(u) = \infty$. So the dot product is admissible at u only if its score never grows in any direction of the future, and nothing in training enforces that. A bounded output layer on g (such as* tanh*) makes the score admissible, by Proposition 7, without normalising anything. Section 5 shows this is not enough in practice.*

Admissibility is not a technicality. It is exactly the condition for the bank forecast to work.

Proposition 10 (Admissible ⇔ the bank forecast converges)**.** *Fix u and draw the bank i.i.d. from μ_v. If $Z(u) < \infty$, then $\nu^N_{v|u} \to \nu_{v|u}$ weakly, almost surely, and $\mathrm{ESS}/N \to 1/\mathbb{E}_{\mu_v}[(d\nu_{v|u}/d\mu_v)^2]$ when that moment is finite. If $Z(u) = \infty$, the normaliser $\frac{1}{N}\sum_j e^{s(u,v_j)}$ diverges and there is nothing to converge to.*

Proposition 11 (Collapse)**.** *If the scores $X_j = s(u, v_j)$ of the bank futures have a regularly varying right tail, $P(X > x) = x^{-\alpha}L(x)$, then $\max_j w_j \to 1$ in probability as $N \to \infty$. This covers the linear dot product on Student-t futures, with α equal to the degrees of freedom.*

So the thesis's two outcomes are, precisely: with a bounded score the bank forecast *converges to the model* as the bank grows; with an unbounded score on heavy tails it *collapses onto the single most extreme future*, and a bigger bank only supplies more extreme futures to collapse onto. Table 2 checks this for fixed scores, with no training.

bank size N	linear score, Gaussian	linear score, Student-t ($\nu = 3$)	L^2 score, Student-t ($\nu = 3$)
10^2	0.081 (0.41)	0.240 (0.30)	0.018 (0.79)
10^3	0.016 (0.38)	0.455 (0.038)	0.002 (0.79)
10^4	0.003 (0.37)	0.760 (0.0007)	$< 10^{-3}$ (0.79)
10^5	0.001 (0.37)	0.902 (10^{-5})	$< 10^{-4}$ (0.79)
10^6	$< 10^{-4}$ (0.37)	0.954 (10^{-6})	$< 10^{-5}$ (0.79)

Table 2: Largest bank weight (mean over 200 banks), with ESS/N in brackets, for a fixed score $s = v$ or $s = -\frac{1}{2}(v-1)^2$ on unit-variance futures. For the Gaussian, theory gives ESS/$N \to e^{-1} = 0.368$.

Proposition 12 (Bounded on both sides: guaranteed bank, capped sharpness)**.** *If $|s| \le \kappa$ (normalised cosine), then $e^{-2\kappa} \le d\nu_{v|u}/d\mu_v \le e^{2\kappa}$. Hence $D_{\mathrm{KL}}(\nu_{v|u}\|\mu_v) \le 2\kappa$ and $\mathrm{ESS}/N \ge e^{-2\kappa}$ in the limit, uniformly in u. The price is that no forecast can be more than $e^{2\kappa}$ times as concentrated as climatology, or carry more than 2κ nats of information.*

	unbounded (dot product)	bounded above (L^2)	bounded both sides (norm. cosine)
model exists	only with light-tailed futures	always	always
bank forecast converges	only with light-tailed futures	always	always
bank needed	can be infinite (collapse)	finite, $\approx e^{D_{\mathrm{KL}}(\nu_{v\|u}\|\mu_v)}$ [13]	uniformly $\lesssim e^{2\kappa}$
sharpness	unlimited	unlimited	capped by κ

Table 3: The boundedness hierarchy.

A subtlety: why training does not avoid the problem. At the population level an inadmissible parameter has infinite loss, so an exact minimiser would avoid it. But training uses the empirical loss, which is always finite: a batch of 256 futures never contains the extreme futures that make $Z(u)$ infinite. So training can settle on an informative parameter whose score grows in the future, and the problem only shows up when the forecast meets a large bank. For the linear dot product on Student-t data, Proposition 8 says every informative parameter is of this kind.

5 Testing the tails directly

We ran two experiments, both on the heavy-tailed domain ($\nu = 3$), with the Gaussian process as a control.

Setup. Each model is trained exactly as in Section 2 (same data, 15k steps, batch 256), then frozen. Only the bank changes: the forecast is rebuilt from held-out banks of $N = 100$ up to $N = 200{,}000$ futures, with up to 3 disjoint banks per size, on the same 500 test pasts. We record the energy score above the perfect forecaster, the largest bank weight, and, for collapsed forecasts (largest weight > 0.5), how extreme the winning future is: its percentile by size within the bank. 2 seeds on the heavy-tailed domain, 1 for the control. The script is `test_bed/analysis/bank_sweep.py`.

Experiment 1: does a bigger bank help or hurt? Theory (Propositions 10, 11) predicts that bounded scores improve and level off as the bank grows, and that the unbounded dot product turns around and gets worse.

	model	$N = 100$	1k	3k	30k	100k	200k
heavy tails, MLP	dot product (unbounded)	+33.6	+8.7	**+7.1**	+17.5	+34.5	+46.3
	dot, tanh future encoder	+37.0	+9.7	+6.7	**+3.2**	+5.4	+7.5
	L^2 one-way	+36.8	+11.8	+8.0	+4.5	+3.8	+2.9
	L^2 joint	+31.5	+6.2	+3.2	+0.9	+0.7	+0.6
	normalised cosine	+41.8	+8.0	+3.5	+1.0	+0.8	+0.8
heavy tails, linear	dot product (unbounded)	**+33.2**	+38.7	+70.2	+1253	+2705	+3358
	dot, tanh future encoder	+28.0	**+22.1**	+24.5	+64.1	+92.3	+102.6
	L^2 one-way	+22.8	+8.1	+5.8	+4.6	+4.3	+4.2
	L^2 joint	+21.5	+7.8	+5.7	+4.5	+4.5	+4.5
	normalised cosine	+25.4	+23.5	+23.2	+23.1	+23.0	+23.1
Gaussian (control), MLP	dot product (unbounded)	+13.7	+2.9	+1.5	+2.0	+1.8	+1.8
Gaussian (control), linear	dot product (unbounded)	+9.9	+6.4	+6.1	+5.8	+5.7	+5.7

Table 4: Energy score above the perfect forecaster (%) as the bank grows, for frozen models. Mean of 2 seeds (control: 1 seed). Bold marks the best bank size where a curve turns around. Full tables: `test_bed/results/bank_sweep/summary.md`.

The prediction holds. On heavy tails the unbounded dot product is best at a small bank and then gets steadily worse: with MLP encoders from +7% at 3k futures to +46% at 200k, and with linear encoders from +33% to over +3000%. Every bounded score improves and levels off. On the Gaussian control the same unbounded score also improves and levels off (at +5.7% linear, matching the +5.6% of Finding 1), so the turnaround is caused by the tails, not by the score alone. The collapse diagnostic shows the mechanism directly. With linear encoders, 20% of the dot product's forecasts at $N = 200$k put more than half their weight on one future, and that future is the most extreme in the bank (99.998th percentile by size, seed 1).

Experiment 2: is a ceiling enough? Proposition 9 says that a tanh on the future encoder makes the dot product admissible without normalising anything. The result is only a partial fix (Table 4, second row of each block):

- it is far better than the uncapped dot product (+103% against +3358% with linear encoders, +7.5% against +46% with MLPs at 200k);
- but it still turns around: in one MLP seed it improves to +1.2%, in the other it rises from +8% to +14%; with linear encoders it rises from +22% to +103%;
- its forecasts rarely collapse onto one future, but the futures they favour are still the most extreme in the bank (99.99th percentile, linear, seed 1).

This does not contradict the theory. The ceiling makes the model exist, so the bank forecast converges. But it converges to a model that over-weights extremes. A tanh score rises *towards* its ceiling as the future becomes more extreme, so extreme futures in the favoured direction receive the top score. A small bank contains few of them and hides the problem, and a large bank exposes it.

What the two experiments say together. The scores that work either *penalise* extreme futures (L^2: the score falls as $g(v)$ moves away from $f(u)$) or are *indifferent* to how extreme a future is

(normalised cosine depends only on the direction of $g(v)$). It is not the height of the ceiling that matters: normalised cosine's learned scale is about 72, so its ceiling is enormous, and it is the best model here. The underlying reason is that training cannot correct the tails. A batch of 256 futures almost never contains one from the far tail, so how the score treats extreme futures is set by the architecture, not learned from data. A ceiling is *necessary*: without one the model does not exist. In practice the score must also not *reward* extremity.

6 Finite data: the loss matters again, for a different reason

The thesis is a statement about infinite data. With finite data the loss variant can matter again, through over-fitting rather than through what it aims at. The clearest case is weather, where one-way L^2 is the well-specified variant yet loses to joint:

skill (2 seeds)	L^2 1-way	L^2 2-way	L^2 joint
rain, linear (vs climatology)	−0.106	+0.011	+0.012
temperature, MLP (vs persistence)	+0.152	+0.243	+0.280
temperature, linear (vs persistence)	+0.352	+0.344	+0.383

The population argument bounds what joint and two-way L^2 can lose to rigidity: in the scalar Gaussian, at most about 3/8 (joint) or 1/3 (two-way) of $I(u;v)$. On rain $I(u;v)$ is tiny (the best model beats climatology by 1%), so that cost is negligible, and the observed gaps must come from finite data. One-way L^2 has nothing that stops it from becoming over-sharp, and on rain its forecasts collapsed (minimum ESS 1.6 with linear encoders, median ESS 3 with MLPs). Two-way and joint L^2 do have such a check: by a Schur complement, $S \succeq A^\top R^+ A$, so a sharp forecast must come with a sharp backward conditional, which the two-way loss penalises; the joint loss also scores the model's marginals, which a collapsed forecast distorts. On this reading the joint loss's small bias buys a regulariser against over-sharpening. It predicts that the one-way deficit shrinks as training data grow, as it does on the plentiful synthetic data of Finding 2. This is a hypothesis, not a result.

7 Relation to the literature

A literature sweep (about fifteen targeted searches, October 2026) found the following relatives. It was not exhaustive, so "we did not find" below means only that.

What is known, and what we build on.

- **The probabilistic framework** is that of Baptista, Stuart & Tran [1]. Stuart & Wolf [2] take the same population-level view and prove that two-tower CLIP is a universal approximator in KL on compact domains. Their extension to unbounded domains requires the score to be bounded outside a compact set, with an error term that grows like e^B in that bound. They also bound the conditional KLs by the joint KL, but do not claim that the loss variants share minimisers. Our first half (the loss stops mattering) is their expressivity result combined with Proposition 1. Our second half explains why their boundedness condition is not just a convenience of the proof.
- **The optimal score is the log density ratio.** That the InfoNCE optimum is the pointwise mutual information plus a function of the conditioning variable is standard (CPC [3]; Poole et al. [4]). For CLIP's symmetric loss, Uesaka et al. [5] show that the optimal similarity is the pointwise mutual information and argue that inner products of finite-dimensional embeddings

may fail to reach it. They propose a richer similarity. That is the closest relative of our misspecification view, but it concerns a single loss, not the agreement or disagreement between variants.

- **Sampling by reweighting candidates.** Sordoni et al. [6] note that resampling candidates with softmax-normalised critic scores approximates $p(y \mid x)$, and that it is exact as the candidate pool grows. That is our bank forecast. They do not study when the limit fails to exist.
- **Bounded critics and normalisation.** In mutual-information estimation, Song & Ermon [7] clip the critic to tame the variance of partition-function estimates. In representation learning, normalised embeddings are standard because they remove the scale degree of freedom that lets the softmax become arbitrarily sharp [8, 9]. Negative squared Euclidean distance logits (our L^2) are used in practice and match or beat cosine CLIP [10]. None of these ties boundedness to the *existence* of the model or to the behaviour of a reweighted bank.
- **Importance-sampling degeneracy.** Weight collapse is classical: in high dimensions it is studied by Bengtsson, Bickel & Li [11], heavy-tailed weights by Pareto-smoothed importance sampling [12], and the required sample size $\approx e^{D_{\mathrm{KL}}}$ by Chatterjee & Diaconis [13]. That an exponential tilt $e^{\langle a,v\rangle}$ needs a finite moment generating function is textbook. Propositions 10 and 11 apply these facts to contrastive scores; we claim no new probability.
- **Analog forecasting.** Forecasting by weighting historical futures is the analog ensemble of weather forecasting [14] and kernel analog forecasting [15]. A contrastive bank forecast is an analog forecast whose kernel, $e^{s(u,v)}$, is learned.

What we did not find.

- The statement that the one-way, two-way and joint losses agree exactly when the score family can represent $\log r$, with the one-way loss needing strictly less (Proposition 1, Corollary 3).
- That linear L^2 is too rigid for the two-way and joint losses on any correlated Gaussian, with the closed form and the 3/8 limit (Propositions 4, 5).
- That a score growing in the future has no valid model on heavy-tailed data even with ReLU encoders, and that its bank forecast then collapses and gets worse as the bank grows (Propositions 8–11, Section 5).
- The thesis itself: with flexible encoders the choice of loss drops out and the choice of score survives only through its tails.

8 Scope and caveats

- Sections 3 and 4 are population-level statements (infinite data, infinite negatives). The finite-sample statements are Propositions 10 and 11, the experiments of Section 5, and the hypothesis of Section 6.
- The information losses in Table 1 are KL divergences; the experiments use the energy score, which is less sensitive to spread errors. The two should agree in sign, not in size.
- Proposition 9 is proved for ReLU networks. Our MLPs use GELU, which grows like ReLU at infinity, so the same conclusion holds, but we have not written out that proof.

- Rain is physically bounded, so its true $Z(u)$ is finite and Proposition 11 does not apply literally. What applies is its finite-N version: a tiny ESS/N whenever the scores over the bank are heavy-tailed.
- All experiments use 2 seeds, except the Gaussian control of Section 5 (1 seed). The collapse diagnostic was recorded for seed 1 only.
- The "does not reward extremity" refinement of Section 5 is an empirical reading of two experiments on one domain. We have not stated or proved it as a theorem.
- Untested so far: the weak-dependence Gaussian prediction of Proposition 5 at finite data, the quadratic-augmented dot product (Table 1, last column) in training, and the bank sweep on rain.

A Proofs

Proposition 1. (i) $J_{v|u}(s) = 0$ iff $\nu_{v|u}(\cdot \mid u) = \mu_{v|u}(\cdot \mid u)$ for a.e. u. Comparing densities with respect to μ_v, that is $e^{s(u,v)}/Z(u) = r(u,v)$, i.e. $s = \log r + \log Z(u)$. Conversely, $s = \log r + c(u)$ gives $Z(u) = e^{c(u)} \int r(u,\cdot)\, d\mu_v = e^{c(u)}$ and hence the true conditional. (ii) is symmetric. (iii) With $\lambda_u, \lambda_v > 0$, $J_2 = 0$ iff (i) and (ii) both hold, so $c(u) = c'(v)$ for $\mu_u \otimes \mu_v$-a.e. (u,v). A function of u alone that equals a function of v alone a.e. under a product measure is a.e. constant. For the joint loss, $D_{\mathrm{KL}}(\mu\|\nu) = 0$ iff $e^s/Z = r$, i.e. $s = \log r + \log Z$. □

Proposition 4. The model joint has density proportional to $e^s\,\mathcal{N}(0, D)$; with $s = -\frac{1}{2} z^\top Q z$, $z = (u, v)$, that is $\mathcal{N}(0, (D^{-1} + Q)^{-1})$. Since $D^{-1} + Q \succeq D^{-1}$, inverting reverses the order, so $(D^{-1} + Q)^{-1} \preceq D$. On the data side, $D - C = \left(\begin{smallmatrix} 0 & -C_{uv} \\ -C_{uv}^\top & 0 \end{smallmatrix}\right)$ has eigenvalues $\pm\sigma_i(C_{uv})$, so $C \preceq D$ forces $C_{uv} = 0$. Zero two-way or joint loss requires the model joint to equal μ (Proposition 1(iii)), which is impossible when $C_{uv} \neq 0$. The inequality $S \succeq A^\top R^+ A$ is the generalised Schur criterion for $\left(\begin{smallmatrix} R & -A \\ -A^\top & S \end{smallmatrix}\right) \succeq 0$. □

Proposition 5. With $s = -\frac{1}{2}(gu - hv)^2$ and unit variances, $Q = \left(\begin{smallmatrix} g^2 & -gh \\ -gh & h^2 \end{smallmatrix}\right)$ and, up to constants, $2J_{\text{joint}} = g^2 + h^2 - 2gh\rho - \log(1 + g^2 + h^2)$. For fixed $t = g^2 + h^2$ the cross term is minimised at $g = h$, leaving $t(1-\rho) - \log(1+t)$, minimised at $t = \rho/(1-\rho)$. The forecast then has precision $1 + t/2 = (2-\rho)/(2(1-\rho))$ and mean coefficient $\rho/(2-\rho)$. Expanding for small ρ gives $J_{v|u} = \frac{3}{16}\rho^2 + O(\rho^3)$ against $I(u;v) = \frac{1}{2}\rho^2 + O(\rho^4)$, so the ratio tends to $3/8$. At $\rho = 0.5$ the closed form gives 0.1699, matching Table 1. □

Proposition 7. $Z(u) = \int e^s\, d\mu_v \leq e^c$, and likewise for Z. For L^2, $-\frac{\kappa}{2}|f - g|^2 \leq 0$; for normalised cosine, $|\langle \bar{f}, \bar{g} \rangle| \leq 1$ by Cauchy–Schwarz. □

Proposition 8. $Z(u) = e^{b(u)} \mathbb{E} \exp\langle H^\top a(u), v \rangle$ by definition. A multivariate t is elliptical, so for $w \neq 0$ the projection $\langle w, v \rangle$ is a scaled univariate t, whose density decays polynomially; hence $\mathbb{E} e^{\langle w, v \rangle} = \infty$. □

Proposition 9. Write $g = W_L \sigma(\dots \sigma(W_1 v + b_1) \dots) + b_L$ with $\sigma = \mathrm{ReLU}$. ReLU is 1-Lipschitz and positively homogeneous, so by induction over layers $|g(v) - g_\infty(v)| \leq \sum_\ell \|b_\ell\| \prod_{k>\ell} \|W_k\| =: B < \infty$, and $g_\infty(tv) = t g_\infty(v)$ for $t \geq 0$. Let $\delta = \langle f(u), g_\infty(v_0) \rangle > 0$. By continuity there is an open cone

$K \ni v_0$ on which $\langle f(u), g_\infty(v/|v|)\rangle \geq \delta' > 0$, so on K, $s(u,v) \geq \kappa(\delta'|v| - |f(u)|B)$. An elliptical t gives every open cone a polynomial tail, $\mu_v(K \cap \{|v| > t\}) \asymp t^{-\nu}$, so $\int_K e^{\kappa\delta'|v|}\, d\mu_v = \infty$. □

Proposition 10. For bounded continuous φ, the strong law gives $\frac{1}{N}\sum_j e^{s(u,v_j)}\varphi(v_j) \to \int e^s \varphi\, d\mu_v$ and $\frac{1}{N}\sum_j e^{s(u,v_j)} \to Z(u)$ almost surely, since both integrands are integrable when $Z(u) < \infty$. The ratio converges to $\int \varphi\, d\nu_{v|u}$; a countable convergence-determining class gives weak convergence. With $\rho = e^s/Z(u)$, $N\sum_j w_j^2 = \frac{\frac{1}{N}\sum \rho(v_j)^2}{(\frac{1}{N}\sum \rho(v_j))^2} \to \mathbb{E}_{\mu_v}[\rho^2]$. If $Z(u) = \infty$, the strong law for nonnegative variables gives $\frac{1}{N}\sum_j e^{s(u,v_j)} \to \infty$. □

Proposition 11. Let $X_{(1)} \geq X_{(2)}$ be the two largest scores and b_N the $1 - 1/N$ quantile of X, so $b_N = N^{1/\alpha}\ell(N)$ with ℓ slowly varying. By the extreme-value limit for regularly varying tails, $(X_{(1)}, X_{(2)})/b_N$ converges in distribution to $(\Gamma_1^{-1/\alpha}, \Gamma_2^{-1/\alpha})$, the first two points of a Poisson process, which are a.s. distinct. So for every $\varepsilon > 0$ there is $c > 0$ with $P(X_{(1)} - X_{(2)} \geq cb_N) \geq 1 - \varepsilon$ for large N, and on that event $1 - \max_j w_j \leq \sum_{j\neq(1)} e^{X_j - X_{(1)}} \leq Ne^{-cb_N} \to 0$, because b_N grows polynomially. □

Proposition 12. $|s| \leq \kappa$ gives $Z(u) \in [e^{-\kappa}, e^{\kappa}]$, so $d\nu_{v|u}/d\mu_v = e^s/Z(u) \in [e^{-2\kappa}, e^{2\kappa}]$. Then $D_{\mathrm{KL}}(\nu_{v|u}\|\mu_v) = \mathbb{E}_\nu \log(d\nu/d\mu_v) \leq 2\kappa$ and $\mathbb{E}_{\mu_v}[(d\nu/d\mu_v)^2] = \mathbb{E}_\nu[d\nu/d\mu_v] \leq e^{2\kappa}$; the ESS limit follows from Proposition 10. □

Reproducibility. Table 1 and Table 2 are produced by `test_bed/theory/misspecification.py`; the experiments of Section 5 by `test_bed/analysis/bank_sweep.py`.

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